Computational Prediction of COVID-19 Risky Genes Associated with Lung Cancer

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Introduction

It has been noticed that people with pre-existing medical conditions who are immunocompromised due to cancer treatment (e.g., chemotherapy) are prone to being infected with the COVID-19 virus and develop severe complications.

There are barely any studies so far that have been conducted about how genes associated with Lung Cancer could also serve as candidate targets of COVID-19.

Along with that, I have heard that Lung Cancer patients with COVID-19 had experienced symptoms way more severe than the symptoms of the normal population.

What am I exactly trying to find or do?

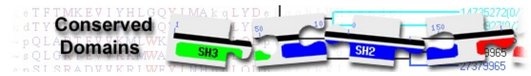
With this research project, I wanted to find the connection between the two diseases, COVID-19 and Lung Cancer, by identifying candidate genes and their molecular mechanisms (e.g. expression patterns).

Methods and Databases

Expression Calculation for Identified Candidate Genes Associated with The Cancer Genome Atlas (TCGA) Lung Adenocarcinoma (LUAD)

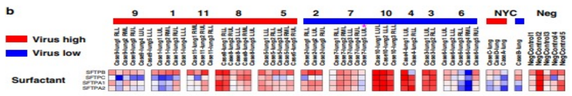
Protein-Protein Interaction (PPI) Network Analysis for Selected Candidate Genes

Functional Annotation and Pathway Analysis for Selected Candidate Down-Regulated Genes Expressed in LUAD



Identification of Conserved Domains for Down-Regulated Genes in LUAD

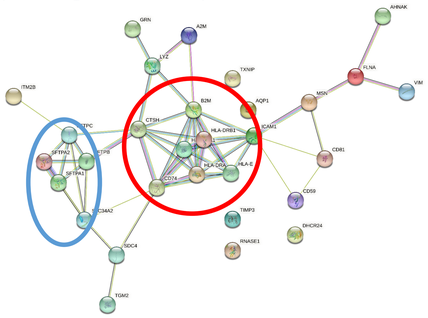
Survival Analysis of Down-Regulated Genes Expressed in LUAD



Cross-check for Down Regulated Genes Expressed in LUAD using SARS-CoV-2 Infection Studies and Literatures

Results and Findings

After conducting PPI (Protein– Protein Interaction for 30 Most Down – regulated Candidate Genes, I found that the genes involved in basic lung functions and the immune deficiency genes interacted directly closely (See figure below).



Functional Annotation Analysis Certified that Genes with a close PPI were reported to have a role in immune functions (See table below).

Category	Term	Genes	P-Value	FDR
INTERPRO	IPR003006:Immunoglobulin/major histocompatibility complex, conserved site	*HLA-DRA, B2M, HLA-DRB1, HLA-DPA1, HLA-E*	2.21E-06	1.92E-04
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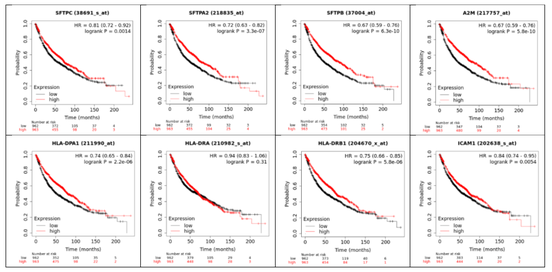
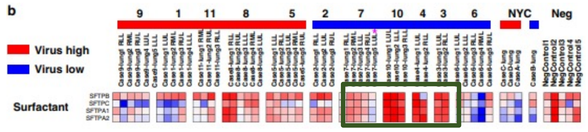
INTERPRO	IPR003597:Immunoglobulin C1-set	*HLA-DRA, B2M, HLA-DRB1, HLA-DPA1, HLA-E*	3.65E-06	1.92E-04
KEGG_PATHWAY	hsa05320: Autoimmune thyroid disease	*HLA-DRA, HLA-DRB1, HLA-DPA1, HLA-E*	4.92E-04	0.001750293
UP_KEYWORDS	Immunity	*CD74, HLA-DRA, B2M, HLA-DRB1, HLA-DPA1, HLA-E*	6.08E-04	0.005251653
GOTERM_BP_DIRECT	GO:0006955~immune response	*CD74, HLA-DRA, B2M, HLA-DRB1, HLA-DPA1, HLA-E*	6.99E-04	0.032497849
GOTERM_BP_DIRECT	GO:0050776~regulation of immune response	*CD81, B2M, ICAM1, HLA-E*	0.003494336	0.129989287
INTERPRO	IPR007110: Immunoglobulin-like domain	*HLA-DRA, B2M, HLA-DRB1, ICAM1, HLA-DPA1, HLA-E*	0.006327346	0.094910188
KEGG_PATHWAY	hsa04672: Intestinal immune network for IgA production	*HLA-DRA, HLA-DRB1, HLA-DPA1*	0.00883398	0.018845825

Looking at the graphs of survival analysis of the down-regulated genes that are expressed in LUAD, I can see that the patients expressing most of the down-regulated genes (22 out of 25) with higher expressions in normal conditions survive l(median) longer.



The low – viral cases had several of our identified surfactant genes (SFTPB, SFTPC, SFTPA1, SFTPA2) that are enriched compared to high viral cases.



Conclusions

After analyzing the results given by the databases we used, I can conclude that when the surfactant genes are down – regulated or under – expressed in patients with Lung Cancer, it makes it easier for COVID-19 to cause harm to the patient. This indicates the likelihood of the co – occurrence of Lung Cancer and Coronavirus Disease. To understand this easier, let's take a city, for example. The patient's body will be the city, The surfactant genes will be the police, the current crimes happening in the city is Lung Cancer, and potential intruders will be COVID-19. The city functions as normal, but with more crimes happening than usual (AKA lung cancer). This causes the police to be busy with managing those crimes and are not as available to protect outside intruders from coming in. This makes it much easier for the intruders to invade and wreak havoc in the city, and when that happens, the city struggles to stay together. This analogy can be implemented to the relationship between Lung Cancer and COVID-19. With Lung Cancer, the patient's surfactant genes become down-regulated, and when COVID-19 invades, the surfactant genes struggle to protect the body. The surfactant genes could serve as potential biomarkers, which will guide future vaccine/drug design.

DRUG	DRUG NAME	DRUG DESCRIPTION
DB15656	**AstraZeneca COVID-19 Vaccine**	A two-dose replication-deficient adenovirus vector vaccine used to prevent COVID-19 caused by SARS-CoV-2.
DB11817	**Baricitinib**	A Janus kinase inhibitor used to treat moderate to severe rheumatoid arthritis that has responded poorly to at least one TNF antagonist.
DB15654	**Moderna COVID-19 Vaccine**	An mRNA vaccine for the prevention of COVID-19, the disease caused by the SARS-CoV-2 virus, in patients aged 18 years and older.
DB15696	**Pfizer-BioNTech COVID-19 Vaccine**	An mRNA vaccine for the prevention of COVID-19, the disease caused by the SARS-CoV-2 virus.
DB14761	**Remdesivir**	A nucleoside analog used to treat RNA virus infections including COVID-19.

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Acknowledgements

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